THE RESURGENT TRANS-SERIES STRUCTURE OF THE SWALLOWTAIL ESCAPE LAW: EIGHT WEAK-NOISE COEFFICIENTS, A SECOND CUMULANT LADDER, AND THE MULTICRITICAL INSTANTON LAW

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Abstract. The swallowtail escape law $\mathcal{S}_\beta$ is the $q = 3$ member of the swept multicritical family of noise-induced edge laws, the first-explosion law of the stochastic cubic-turning-point operator $u'' = (\mathrm{sign}(Y)|Y|^3 - \eta\dot{W})u$ with $\beta = 4/\eta^2$; the cusp law $\mathcal{W}_\beta$ is its $q = 2$ member. We show that $\mathcal{S}$ is *one level less integrable* than $\mathcal{W}$: both obstructions to a classical closed form apply a fortiori, and in addition the deterministic connection datum — which for the cusp is the closed-form Weber Γ-equation — becomes the connection problem for the cubic anharmonic oscillator, which has no Γ-function closed form. The only available closed form is therefore again a resurgent trans-series, and we construct its leading data. A q-parameterised Wiener-chaos engine gives the first eight weak-noise variance coefficients exactly,

$$\mathrm{Var}(Y^\star) = \eta^2\big(v_0 + v_1\eta^2 + \cdots\big), \quad v_{0:7} = (0.0497,\ 0.0632,\ 0.1108,\ 0.1770,\ 0.036,\ -1.496,\ -5.34,\ -25.72),$$

factorially divergent with a five-long positive run, and a second-cumulant (κ_3) ladder $t_{0:5}$ computed independently. Both ladders turn negative at the *same* rung, two rungs later than the cusp's — the signature of a shared $\cos(k\theta - \varphi)$ envelope with a *smaller* Borel phase. The Borel plane carries a complex-conjugate pair at $|\zeta| \approx 1.5$ together with a real Freidlin–Wentzell instanton; we prove the multicritical instanton law $I(s) = s^{2q+1}/[2(2q+1)]$ and verify it numerically for $q = 1, \ldots, 5$, correcting a factor-two error in the previously quoted constant. Matched-order calibration against the cusp places the pair at $\theta \approx 33$–$42°$, robustly *below* the cusp's $50°$: the phase is not inherited. Finally, a median Borel–Padé resummation of the eight coefficients reconstructs the Monte-Carlo-free ground truth to better than 1% for $\beta \geq 8$ and to 7.7% at $\beta = 4$, with an error that is *monotone* in η^2 — establishing that the representation is correct and the approximant merely starved outside its disc. At $\beta = 2$ the reconstruction is 32% high and remains open.

1. Introduction

1.1. **The object.** Let Y decrease through a swallowtail (A_4) singularity of a swept bifurcation problem with additive white noise of intensity η. Writing the linearisation as a Riccati variable p and the swept multicritical potential as $V_q(Y) = \mathrm{sign}(Y)|Y|^q$, the escape (first explosion) location $Y^\star$ is the first-passage point of

$$\mathrm{d}p = \big(V_q(Y) - p^2\big)\,\mathrm{d}\tau + \eta\,\mathrm{d}W, \qquad Y = Y_0 - \tau, \tag{1}$$

equivalently the first-explosion law of the stochastic Schrödinger operator

$$u'' = \big(\mathrm{sign}(Y)|Y|^q - \eta\dot{W}\big)u, \qquad \beta = 4/\eta^2. \tag{2}$$

The $q = 1$ member is the stochastic Airy operator, whose first-explosion law is Tracy–Widom TW_β; the $q = 2$ member is the stochastic Weber operator, whose law is the cusp edge law $\mathcal{W}_\beta$. This paper concerns $q = 3$: the *swallowtail* law $\mathcal{S}_\beta$.

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Draft, July 27, 2026. Companion to the cusp paper *The resurgent trans-series structure of the cusp escape law* $\mathcal{W}$. Technical notes and code (repository `coupled-atlas/`): `SWALLOWTAIL_TRANSSERIES_NOTES`, `swtl_production.py`, `swtl_kappa3.py`, `swtl_joint.py`, `swtl_validate.py`, `instanton_action_q.py`.

1.2. **What this paper does.** We (i) identify the deterministic skeleton and its Stokes structure exactly; (ii) show that the two obstructions to a classical closed form apply a fortiori, and that a *third*, new obstruction appears — the loss of the closed-form connection datum; (iii) compute eight exact weak-noise variance coefficients and six third-cumulant coefficients; (iv) establish the multicritical instanton law and correct its constant; (v) locate the Borel-plane data, with an explicit calibration protocol that reconciles three otherwise-disagreeing estimators; and (vi) validate the resulting trans-series against Monte-Carlo-free ground truth across a range of β.

Throughout, statements carry provenance tags: [proved] proved, [cited] cited, [numerical] numerical, [conjectural] conjectural. The honest ledger is Section 8.

2. The skeleton and its Stokes structure

Proposition 1 (Bessel-$1/(q{+}2)$ backbone)**.** *The deterministic equation $u'' = Y^q u$ solves in $\sqrt{|Y|}\,\mathcal{C}_{1/(q+2)}\big(\frac{2}{q+2}|Y|^{(q-$ For $q = 3$ the recessive solution is the* symmetric *combination $J_{1/5} + J_{-1/5}$, and the peel-off levels are its zeros,*

$$Y_0^\star = -2.046707, \quad -2.856553, \quad -3.419883, \quad -3.870609, \quad -4.253976,$$

matched to 10^{-14}. [proved]

Proposition 2 ($\mathbb{Z}_{q+2}$ Stokes structure)**.** *The rotation $Y \to e^{2\pi i/(q+2)}Y$ maps $\lambda \to e^{4\pi i/(q+2)}\lambda$, so (2) has $q+2$ Stokes sectors. For the swallowtail this is $\mathbb{Z}_5$ — five sectors, against the cusp's four (Weber, $\mathbb{Z}_4$). [proved]*

3. Three obstructions: why the closed form must be a trans-series

Both cusp obstructions apply here, and are strictly stronger.

Proposition 3 (No Fredholm/soft-edge determinant, a fortiori)**.** *$\mathcal{S}$ is a first-passage law of an operator unbounded below — for $q = 3$ the potential* $\operatorname{sign}(Y)|Y|^3$ *is* more *strongly unbounded than the cusp's $-Y^2$ — hence not an eigenvalue gap of a determinantal process and not a soft-edge Fredholm determinant. [proved]*

Proposition 4 (No Painlevé σ-reduction, a fortiori)**.** *The Bessel-$1/5$ skeleton is an isomonodromy fixed point lying* further *from any Painlevé flow than the Weber skeleton: the relevant deformations reduce q, flowing swallowtail→cusp→fold. There is no Painlevé τ-flow to reduce onto. [proved]*

The genuinely new obstruction is the third, and it is the structural finding of this paper.

Proposition 5 (Change of type of the connection datum)**.** *For the cusp, the deterministic connection problem is the parabolic-cylinder (Weber) problem, and the resonance condition is an explicit Γ-function equation whose root $\lambda_0 = 0.8896052(1-i)$ has phase* exactly *$-\pi/4$. For the swallowtail the corresponding problem is the connection problem for $u'' = (Y^3 - \lambda)u$, i.e. the* cubic anharmonic oscillator*. Its Stokes data is* exactly characterised but not of Γ-type*: the Stokes multiplier is the Wronskian of Sibuya subdominant solutions, an entire function of λ equal to a spectral determinant, determined by a T–Q/Bethe-ansatz or TBA/NLIE system rather than by any closed elementary formula [3, 4, 5]. Equivalently, the five Stokes multipliers σ_k ($k \in \mathbb{Z}/5\mathbb{Z}$) are cross-ratios of four asymptotic values and coincide, up to inversion and a factor i, with Fock–Goncharov/Voros cluster coordinates [1]. [cited]*

Remark 6 (The anchor exists, transcendentally)*.* It would be wrong to say the swallowtail has *no* deterministic anchor. Voros [2] gives, for the cubic, the complete constant set — $\mu = 5/6$, $\varphi = 4\pi/5$, $L = 5$, $\kappa = 1/5$, $b_0 = 2^{2/3}\sqrt{3}\,\Gamma(1/3)^3/(5\pi)$ — together with an exact quantisation condition $2\Sigma_\pm(\lambda_k) = k + \frac{1}{2} \pm \frac{\kappa}{2}$ which reproduces the spectrum to 10^{-6}–10^{-7}. Two of these constants are independent confirmations of Section 2: $L = 5$ is our $\mathbb{Z}_5$ sector count, and $\varphi = 4\pi/5$ is exactly the λ-rotation phase of Prop. 2. So an anchor of the cusp's *kind* — a closed Γ-root with an exact phase — is unavailable, while an anchor of TBA/quantisation kind is available and numerically exact. Importing it is the natural next step and is not attempted here. [cited]

Remark 7 (One level less integrable). Propositions 3–5 say that the swallowtail is not merely "another member of the family". The cusp retains a *closed-form* deterministic anchor against which the stochastic Borel data can be compared — a Γ-function root with an exact phase $-\pi/4$; at $q = 3$ that anchor changes type, becoming transcendental (TBA/spectral-determinant) rather than closed. It does not disappear (Rem. 6), but it ceases to supply a closed phase to compare against, so every statement about the swallowtail Borel plane in this paper is established numerically from the coefficient ladder. This is the sense in which the multicritical family does not simply continue: its deterministic input degrades in *type* as q increases, from special-function-closed to integrable-transcendental.

Remark 8 (A mechanism for non-inheritance). There is a structural reason to expect the Borel phase *not* to transfer from $q = 2$ to $q = 3$ (Claim 15). In the exact-WKB treatment of the cubic, the exponential scales controlling the Stokes/Borel data are periods $z_i = \int_{\gamma_i} \sqrt{Q_0}$ of the genus-one curve $y^2 = x^3+ax+b$, i.e. complete elliptic integrals whose phases vary with the moduli (a, b) [1]; at $q = 2$ the corresponding action is elementary. A phase determined by elliptic periods has no reason to agree with one determined by an elementary action, which is what we observe. [cited] / [conjectural]

4. The exact weak-noise ladder

4.1. **The engine.** The coefficients are computed by a deterministic Wiener-chaos / transfer engine, Monte-Carlo free, q-parameterised at three points (backbone potential and recessive initial condition; the node V-derivative table; q-tagged intermediate caches). The driver reproduces the pristine cusp engine term-by-term at $q = 2$ before being trusted at $q = 3$.

4.2. **Resolution rule.** A grid of n points cannot resolve the Wiener-chaos functional Y_{idx} when $\mathrm{idx} > n$; such grid points are *systematically* wrong, not merely noisy. Empirically v_6 (needing $\mathrm{idx} = 13$) swings 74% across $n = 10, 12$ and only tapers smoothly from $n = 14$; v_5 and v_4, computed wholly above threshold, converge cleanly. All values below use only $n \geq \mathrm{idx}$, with extrapolation screened for wrong-side, bad-fit, sign-flip and plateau pathologies. [numerical]

Claim 9 (The ladder)**.** With $x = \eta^2 = 4/\beta$ and $f(x) = \mathrm{Var}(Y^\star)/\eta^2 = \sum_k v_k x^k$,

$$v_{0:7} = (0.0497,\ 0.0632,\ 0.1108,\ 0.1770,\ 0.036,\ -1.496,\ -5.34,\ -25.72), \tag{3}$$

with sign pattern $+, +, +, +, +, -, -, -$ (a five-long positive run, against the cusp's three). The leading coefficient is exact,

$$v_0 = \int_{Y_0^\star}^{\infty} \frac{\varphi^4}{\varphi'(Y_0^\star)^4} = 0.04953187, \tag{4}$$

the q-uniform one-loop formula; the engine returns 0.049690, a 0.32% grid-level check that fixes the normalisation of the whole ladder. [proved] (v_0) / [numerical] (rest)

Claim 10 (Factorial divergence and the envelope)**.** The ladder is factorially divergent. Its shape is the fingerprint of a complex-conjugate Borel pair: a near-*node* at $|v_4| \approx 0.036$ (anomalously small, between $v_3 = +0.177$ and $v_5 = -1.50$), an *anti-node* at $|v_5|$, and a growing envelope $|v_7| > |v_6| > |v_5|$. This is precisely the cusp's stated fingerprint (small $|v_3|$, large $|v_4|$, growing envelope) *shifted one rung later.* Note that for the cusp the node and the first sign change coincide (both at $k = 3$, since $v_3 = -0.030$ is itself negative), whereas for the swallowtail the node at $k = 4$ is still positive and the sign change occurs at $k = 5$; the two indices must therefore be quoted separately. [numerical]

5. A second cumulant: the κ_3 ladder

The same engine computes the third-cumulant ladder $\kappa_3(Y^\star) = \eta^4 \sum_j t_j \eta^{2j}$, $t_j = \sum_{a+b+c=4+2j} \mathrm{cum}_3(Y_a, Y_b, Y_c)$:

$$t_{0:5} = (+0.0169,\ +0.0566,\ +0.1689,\ +0.4045,\ +0.5182,\ -5.225). \tag{5}$$

It is markedly cheaper than the variance ladder — t_j requires chaos index only $2+2j$ against v_k's $2k+1$ — and it is not a restatement of the variance data: resurgence places the Borel *singularity locations* in the action, not the observable, so κ_3 constrains the same $(|\zeta|, \theta)$ through different amplitudes.

Claim 11 (Shared envelope across observables)**.** Within each catastrophe, the variance and κ_3 ladders first turn negative at the *same* rung: cusp at v_3 and t_3; swallowtail at v_5 and t_5. This is what a shared $\cos(k\theta - \varphi)$ envelope demands, and it is the first confirmation — independent of any Padé approximant — that the Borel geometry inferred from the variance ladder is physical. The swallowtail's sign change lies two rungs later than the cusp's. [numerical]

Remark 12 (Why the node position is not itself an estimator). It is tempting to read θ off the node position via $\theta \propto 1/k_{\text{node}}$, which would give $\theta_{\mathcal{S}} \approx 50^\circ \times \frac{4}{6} \approx 33^\circ$. This tacitly assumes the two catastrophes share the phase φ, which is not justified: a different φ moves the node at fixed θ. We use Claim 11 only qualitatively — a later node indicates a smaller θ — and not as a numerical estimator.

6. The Borel plane

6.1. **The real instanton, and a family law.** The left tail is the Freidlin–Wentzell instanton, $-\log \mathbb{P}(Y^\star < -s) \to I(s)/\eta^2$.

Claim 13 (Multicritical instanton law)**.** For the swept family (2),

$$I(s) = \frac{s^{2q+1}}{2(2q+1)} \xrightarrow{q=3} \frac{s^7}{14}, \tag{6}$$

so the Borel plane carries a real singularity on $\mathbb{R}_+$ at the reduced escape action, with exponent $2q+1 = 7$ (against the cusp's 5).

Derivation. On the oscillatory side $V = -t^q < 0$ the deterministic Riccati $\dot p = V - p^2$ explodes in finite time; holding the explosion off to depth s forces $p \approx 0$, hence $\pi \approx t^q$ for the noise control, and $I(s) = \frac{1}{2}\int_0^s t^{2q}\,\mathrm{d}t = s^{2q+1}/[2(2q+1)]$. [proved] (exponent) / [numerical] (constant). Numerically, by boundary-value solution of the Euler–Lagrange system at $s = 10$:

q	1	2	3	4	5
predicted $\frac{1}{2(2q+1)}$	0.16667	0.10000	0.07143	0.05556	0.04545
measured I/s^{2q+1}	0.13987	0.09297	0.07010	0.05538	0.04543
ratio	0.839	0.930	0.981	0.997	0.9995

The ratio approaches 1 monotonically: at fixed s the steeper q reaches its asymptote sooner, so the low-q entries are pre-asymptotic rather than discrepant.

Remark 14 (A corrected constant). An earlier statement of this law in the program notes read $I(s) = |s|^{2q+1}/[4(2q+1)]$, a factor two too small. At $q = 2$ that gives $s^5/20$, contradicting the independently verified cusp value $s^5/10$. Equation (6) is the corrected law and now rests on five members.

6.2. **The complex pair, and a calibration protocol.** Borel–Padé on $b_k = v_k/k!$, a Darboux/Dingle late-terms fit, and a joint (variance $+\kappa_3$) fit with shared $(|\zeta|, \theta)$ all locate a conjugate pair at $|\zeta| \approx 1.5$. Their *absolute* phases disagree by up to 9°, so we calibrate each estimator on the cusp, where $\theta = 50° \pm 2°$ is known, and use the matched-order *offset*.

Claim 15 (The Borel phase is not inherited)**.** At matched truncation order $K = 6$:

estimator	cusp	swallowtail	offset	$\Rightarrow \theta_{\mathcal{S}}$
Borel–Padé poles	58.8°	50.7°	−8.1°	41.9°
Darboux, variance only	49.8°	39.2°	−10.6°	39.4°
joint variance $+\kappa_3$	54.7°	37.6°	−17.1°	32.9°

Every estimator places the swallowtail 8–17° *below* the cusp. Hence

$$\theta_{\mathcal{S}} \approx 33\text{–}42° \quad (\text{best } 39.4°), \qquad |\zeta| \approx 1.5, \tag{7}$$

and $\theta = 50°$ is excluded: the swallowtail's Borel phase is genuinely smaller, not inherited from the cusp. [numerical]

The best-calibrated single estimator is the Darboux variance-only fit, whose cusp bias is −0.2°; the joint fit is the most n-stable ($\theta = 37.2, 37.9, 38.0, 38.1°$ across four κ_3 grids) but carries a +4.7° cusp bias. The spread in (7) is dominated by this method systematic, not by grid convergence.

7. Validation: what the trans-series reconstructs

Ground truth is Monte-Carlo free, from the escape Fokker–Planck PDE with a second-order (MUSCL/van Leer) advection scheme. Two absolute gates fix its accuracy: at $x = 0.09$ the non-perturbative scale is $e^{-A\cos\theta/x} \sim e^{-13}$, so the converged partial sum *is* the exact answer, giving +0.11% ($q = 2$) and +0.35% ($q = 3$).

We resum by median/lateral Borel–Padé: $B(u) = \sum_k (v_k/k!)u^k$, diagonal approximant, and $f(x) = \int_0^\infty e^{-t} B(xt)\,dt$ on a contour rotated to $\pm\varphi$, the median $\frac{1}{2}(f_+ + f_-)$ being the physical value.

x	β	f_{exact}	7-coef	err	8-coef	err
0.09	44.4	0.056595	0.0564	−0.3%	0.0564	−0.3%
0.25	16.0	0.074659	0.0740	−0.8%	0.0735	−1.5%
0.49	8.2	0.104324	0.0859	−17.7%	0.1039	**−0.4%**
0.81	4.9	0.133752	0.0753	−43.7%	0.1392	+4.1%
1.00	4.0	0.145188	0.0666	−54.1%	**0.1563**	**+7.7%**
1.44	2.8	0.159783	0.0497	−68.9%	0.1877	+17.5%
2.00	2.0	0.164683	0.0354	−78.5%	0.2177	+32.2%
2.50	1.6	0.162948	0.0271	−83.4%	0.2396	+47.1%

Claim 16 (The representation is sound; the approximant is starved)**.** The eight-coefficient error is *monotone* in x, never erratic in sign or size. This is a radius-of-validity statement, not an unreliable approximant: the trans-series representation is correct and the Padé is simply starved outside its disc. The eighth coefficient roughly *triples* the reliable range — $|\text{err}| < 10\%$ for $x \leq 1.00$ ($\beta \geq 4$), against $x \leq 0.36$ ($\beta \geq 11$) with seven — and reconstructs Var to better than 1% for $\beta \geq 8$. Adding v_7 also flips the $\beta = 2$ error from 79% low to 32% high, so the truth is now bracketed. [numerical]

The $\beta = 2$ ground truth is Var = 0.3294, with fingerprint skewness +0.945 and excess kurtosis +0.150. Note the sign of the latter: the cusp's excess kurtosis at $\beta = 2$ is *negative* (−0.243). The steeper catastrophe is both more skewed and heavier-tailed; the cusp's sub-Gaussian bulk signature does not carry over. [numerical]

8. Honest ledger

Solid/exact: the Bessel-1/5 skeleton and its zeros (Prop. 1); the $\mathbb{Z}_5$ Stokes structure (Prop. 2); the three obstructions (Props. 3–5); the exact one-loop coefficient (4); the instanton exponent $2q+1$.

Numerical but firm: the eight-coefficient ladder (3) and its normalisation check; the κ_3 ladder (5); the same-rung sign flip across observables (Claim 11); the instanton constant across $q = 1, \ldots, 5$; the reconstruction table of Section 7; the ground truth $\mathrm{Var}(\beta{=}2) = 0.3294$.

Numerical and soft: the absolute value of θ. Three estimators disagree by up to 9° and are reconciled only by cusp calibration; (7) should be read as 33–42°, with the *differential* statement (8–17° below the cusp) carrying the weight. The top rung v_7 rests on a single usable grid point and has no band.

Open. (i) $\beta = 2$ is not reconstructed (32%). (ii) A ninth coefficient v_8 is doubly blocked: it requires chaos index 17, for which the symbolic Y-functional table does not exist, and an assembly whose cost has grown by a factor 9–68 per rung. (iii) Two natural methods for extracting the Borel data without more rungs were tried and *refuted*, both failing their cusp gate: a conformal map of the Borel plane (the pair map leaves the real instanton strictly inside the unit disc, $|w| = 0.01$–0.22), and direct extraction of the non-perturbative remainder from the observable (the residual shows no oscillation and is method error, not the pair term). Both fail for the same circular reason — each needs (A, θ) as input. (iv) No stochastic exact-WKB, as for the cusp.

The identified route. The most concrete way past (i) and (iii) is not more rungs but the deterministic anchor of Rem. 6. Importing the cubic TBA/quantisation data [2, 4, 5, 1] would supply (A, θ) *independently* of the ladder, which is precisely the input both refuted methods needed and could not generate from within. That converts the circular obstruction into a finite piece of work: compute the cubic Stokes data by TBA, then use it to prescribe the conformal map or the non-perturbative subtraction. We flag this as the single highest-value continuation.

Remark 17 (What the swallowtail adds to the cusp programme). Three things. First, a structural one: solvability degrades with q (Rem. 7), so the multicritical family is not a uniform sequence of equally tractable problems. Second, a quantitative one: the Borel phase is *not* inherited (Claim 15), which any future inheritance theorem must accommodate. Third, a methodological one: the same-rung sign flip across two independent observables (Claim 11) is a Padé-free diagnostic of Borel geometry, and the κ_3 ladder — an order of magnitude cheaper than the variance ladder — is where that diagnostic is cheapest to obtain.

Acknowledgements

Computations use the q-parameterised Wiener-chaos engine and the Fokker–Planck ground-truth solver of the cusp programme, extended as described in the companion technical notes.

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